COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 8 : SLO 1 — Non-Restoring Division Algorithm
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Section A: Match the Following (Remember)

- A. Non-Restoring Division → 3. No restoration; handled in next cycle
- B. Shift Left Operation → 2. Moves remainder and quotient bits to the left
- C. Negative Partial Remainder → 1. Adds divisor if the remainder is negative
- D. Quotient Bit = 0 → 4. Indicates subtraction failed
- E. Positive Partial Remainder → 5. Indicates subtraction succeeded

Section B: Fill in the Blanks (Understand)

- Non-restoring division removes the need for the RESTORATION step found in restoring division.
- If the partial remainder is negative, the divisor is ADDED in the next step.
- The algorithm uses an initial SHIFT of the remainder and quotient registers.
- The QUOTIENT bit is set to 0 when the partial remainder is negative.
- The division result consists of the QUOTIENT and the final remainder.

Section C: Simulate the Algorithm (Apply): 0100 (4) ÷ 0010 (2)

- Init: A = 0000, Q = 0100, M = 0010
- Step 1: shift (A,Q) → A = 0000, Q = 1000; A ≥ 0 → A = A - M = 1110 (< 0), Q0 = 0 → A = 1110, Q = 1000
- Step 2: shift → A = 1101, Q = 0000; A < 0 → A = A + M = 1111 (< 0), Q0 = 0 → A = 1111, Q = 0000
- Step 3: shift → A = 1110, Q = 0000; A < 0 → A = A + M = 0000 (≥ 0), Q0 = 1 → A = 0000, Q = 0001
- Step 4: shift → A = 0000, Q = 0010; A ≥ 0 → A = A - M = 1110 (< 0), Q0 = 0 → A = 1110, Q = 0010
- Final correction: A < 0 → A = A + M = 1110 + 0010 = 0000
- Final Quotient = 0010 (2), Remainder = 0000 (0) (4 ÷ 2 = 2 remainder 0 ✓)

Section D: Role-Play Debrief (Analyze)

- More efficient than restoring because the extra restore (add-back) step inside every failed cycle is eliminated — each cycle performs exactly one add or subtract, so the loop is shorter and faster in hardware.
- The end correction is needed because if the last partial remainder is negative it does not represent a valid remainder; adding the divisor once restores the true remainder.
- In restoring division the quotient bit is decided after the restore check; in non-restoring the quotient bit is set directly from the sign of the partial remainder (1 if non-negative, 0 if negative) without restoring first.